QUESTION ANALYSIS

EEE421

Solid State Devices

Question Analysis

Sessions 2020–21 *1 papers, 24 questions, 45 formulas*

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 24 questions and 145 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Semiconductor in equilibrium	30	21 %	5	Ionization (15); Fermi level (5); Compensation (5)
Carrier transport	25	17 %	4	Mobility (10); Hall effect (10); Einstein relation (5)
PN junction	25	17 %	4	Depletion width (10); Biased junction (10); Band diagram (5)
Excess carriers	20	14 %	3	Steady state (10); Minority carriers (5); Carrier pulse (5)
Quantum basics	15	10 %	2	Fermi probability (10); Density of states (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Quantum basics	**15**	**15**
Fermi probability	**10**	10
Density of states	**5**	5
Semiconductor in equilibrium	**30**	**30**
Ionization	**15**	15
Fermi level	**5**	5
Compensation	**5**	5
Carrier concentrations	**5**	5
Carrier transport	**25**	**25**
Mobility	**10**	10
Hall effect	**10**	10
Einstein relation	**5**	5
Excess carriers	**20**	**20**
Steady state	**10**	10
Minority carriers	**5**	5
Carrier pulse	**5**	5
PN junction	**25**	**25**
Depletion width	**10**	10
Biased junction	**10**	10
Band diagram	**5**	5
Tunnel diode	**5**	**5**
Operation	**5**	5
MOS structure	**15**	**15**
Band diagrams	**5**	5
Double gate	**5**	5
Inversion	**5**	5
Bipolar transistor	**10**	**10**

Topic and subtopic	2020-21	All
Operating modes	**5**	5
Minority distribution	**5**	5

3. Priority list:

Must know: Semiconductor in equilibrium (30), Carrier transport (25), PN junction (25), Excess carriers (20), Quantum basics (15), MOS structure (15), Bipolar transistor (10), Tunnel diode (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Quantum basics:

15 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 3.

Most often numerical (2 of 2). Subtopics: Density of states; Fermi probability.

Formulas:

Subtopic	Formula	Equation	Used in
Density of states (5)	**Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 1(c)
	States from E_c to $E_c + kT$	$N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$	20-21 1(c)
Fermi probability (10)	Energy for a given probability	$E - E_F = kT \ln\left(\frac{1}{f_F} - 1\right)$	20-21 2(d)

Symbols:

g_c = states per unit energy per unit volume ↩

Questions:

No.	Paper	Marks	Type	Question
Density of states				
1	20-21 1(c)	5	numerical	Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K.
Fermi probability				
2	20-21 2(d)	10	numerical	Answer the following questions on Fermi level: (i) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ? (ii) Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}).

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4.2. Semiconductor in equilibrium:

30 marks, 5 questions, in 1 of 1 papers. Book: Neamen Ch. 4.

Most often theory (2 of 5). Subtopics: Fermi level; Ionization; Compensation; Carrier concentrations.

Formulas:

Subtopic	Formula	Equation	Used in
Fermi level (5)	**Fermi–Dirac function**	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$	20-21 1(a)
	Electron concentration	$n_0 = N_c \exp\left[-\dfrac{E_c-E_F}{kT}\right]$	20-21 1(a)
	Hole concentration	$p_0 = N_v \exp\left[-\dfrac{E_F-E_v}{kT}\right]$	20-21 1(a)
	Mass-action law	$n_0 p_0 = {n_i}^2 = N_c N_v \exp\left(-\dfrac{E_g}{kT}\right)$	20-21 1(a)
	Fermi level vs doping	$E_F - E_{Fi} = kT\ln\dfrac{N_d}{n_i}$ (n-type), $E_{Fi} - E_F = kT\ln\dfrac{N_a}{n_i}$ (p-type)	20-21 1(a)
Ionization (15)	N_c at another temperature	$N_c(T) = N_c(300)\left(\dfrac{T}{300}\right)^{3/2}$	20-21 1(d)
	Fermi level from E_c	$E_c - E_F = kT\ln\dfrac{N_c}{n_0}$	20-21 1(d)
	Donor occupation	$f_d = \dfrac{1}{1+\frac{1}{2}\exp\left(\dfrac{E_d-E_F}{kT}\right)}$	20-21 1(d)
	Fraction not ionized	$\dfrac{n_d}{n_d+n_0} = \dfrac{1}{1+\dfrac{N_c}{2N_d}\exp\left[-\dfrac{E_c-E_d}{kT}\right]}$	20-21 1(d), 20-21 6(b)
	Freeze-out estimate	$n_0 \approx \sqrt{\dfrac{N_c N_d}{2}}\exp\left[-\dfrac{E_c-E_d}{2kT}\right]$	20-21 1(d)
Carrier concentrations (5)	**Charge neutrality**	$n_0 + N_a^- = p_0 + N_d^+$	20-21 3(a)
	Electrons, compensated	$n_0 = \dfrac{N_d-N_a}{2} + \sqrt{\left(\dfrac{N_d-N_a}{2}\right)^2 + n_i^2}$	20-21 3(a)
	Holes, compensated	$p_0 = \dfrac{N_a-N_d}{2} + \sqrt{\left(\dfrac{N_a-N_d}{2}\right)^2 + n_i^2}$	20-21 3(a)

Symbols:

f_F = probability that a state at energy E holds an electron ↩

E_F = Fermi level ↩

N_c = effective density of states in the conduction band ↩

N_v = effective density of states in the valence band ↩

n_i = intrinsic carrier concentration ↩

E_{Fi} = intrinsic Fermi level (near midgap) ↩

E_d = donor level ↩

n_d = electrons still in donor states ↩

N_a^-, N_d^+ = ionized acceptors and donors (cm^{-3}) ↩

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Questions:

No.	Paper	Marks	Type	Question
Fermi level				
1	20-21 1(a)	5	theory	Sketch the position of Fermi level in Si at 300 K as a function of doping concentration.
Ionization				
2	20-21 1(d)	10	numerical	A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner *[sic: should be donor]* states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si].
3	20-21 6(b)	5	derivation	Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer.
Compensation				
4	20-21 2(a)	5	theory	Draw energy band diagram of a compensated semiconductor.
Carrier concentrations				
5	20-21 3(a)	5	derivation	Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i.

4.3. Carrier transport:

25 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 5.

Most often theory (2 of 4). Subtopics: Einstein relation; Mobility; Hall effect.

Formulas:

Subtopic	Formula	Equation	Used in
Einstein relation (5)	**Einstein relation**	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$	20-21 2(b)
Mobility (10)	**Drift velocity**	$v_d = \mu E$	20-21 3(c)
	Mobility	$\mu = \frac{e\tau}{m^*}$	20-21 3(c)
	Conductivity	$\sigma = e(n\mu_n + p\mu_p)$	20-21 3(c)
	Temperature and mobility	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$	20-21 4(a)
Hall effect (10)	**Hall voltage**	$V_H = \frac{I_x B_z}{epd}$ (p-type), $V_H = -\frac{I_x B_z}{end}$ (n-type)	20-21 5(d)
	Carrier density, Hall	$n = -\frac{I_x B_z}{e d V_H}$	20-21 5(d)
	Hall mobility	$\mu_n = \frac{I_x L}{e n V_x W d}$	20-21 5(d)

Symbols:

μ = mobility (cm^2/V s) ↩

E = electric field (V/cm) ↩

τ = mean time between collisions (s) ↩

m^* = conductivity effective mass ↩

μ_L, μ_I = lattice- and impurity-limited mobility ↩

N_I = ionized impurity concentration ↩

d = thickness along B_z

L = length along x, W = width along y

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Questions:

No.	Paper	Marks	Type	Question
Einstein relation				
1	20-21 2(b)	5	derivation	Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation.
Mobility				
2	20-21 3(c)	5	theory	Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ.
3	20-21 4(a)	5	theory	Explain the effects of temperature on carrier mobility inside a semiconductor.
Hall effect				
4	20-21 5(d)	10	numerical	Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. (i) What is the Hall voltage? (ii) What is type of carrier? (iii) What is the carrier concentration? (iv) What is the mobility?

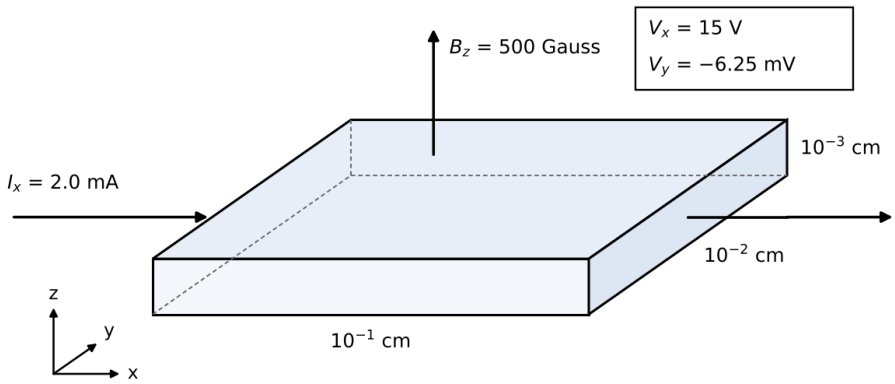


Figure 1: 20-21 5(d): Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

4.4. Excess carriers:

20 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 6.

Most often theory (2 of 3). Subtopics: Minority carriers; Steady state; Carrier pulse.

Formulas:

Subtopic	Formula	Equation	Used in
Minority carriers (5)	**Diffusion current**	$J_n = eD_n\frac{dn}{dx},\ J_p = -eD_p\frac{dp}{dx}$	20-21 1(b)
	Poisson's equation	$\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}$	20-21 1(b)
Steady state (10)	**Ambipolar transport**	$D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$	20-21 3(d)
	Steady-state diffusion	$\delta n(x) = \delta n(0)\exp\left(-\frac{x}{L_n}\right),\ L_n = \sqrt{D_n\tau_{n0}}$	20-21 3(d)
Carrier pulse (5)	n-type, low injection	$D_p\frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0\frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$	20-21 5(a)
	Moving pulse	$\delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}}\exp\left(-\frac{t}{\tau_{p0}}\right)\exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$	20-21 5(a)

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Symbols:

D_n, D_p = diffusion coefficients (cm^2/s) ↩

ϕ = potential (V) ↩

ρ = charge density (C/cm^3) ↩

L_n = minority-electron diffusion length ↩

N_0 = pairs created per unit area at $x = 0$, $t = 0$ ↩

Questions:

No.	Paper	Marks	Type	Question
Minority carriers				
1	20-21 1(b)	5	theory	How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction?
Steady state				
2	20-21 3(d)	10	numerical	For the following questions: (i) Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}). (ii) Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1}) (iii) Plot n and p as function of x in log scale.
Carrier pulse				
3	20-21 5(a)	5	theory	Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time.

4.5. PN junction:

25 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 7, 8.

Most often numerical (2 of 4). Subtopics: Depletion width; Band diagram; Biased junction.

Formulas:

Subtopic	Formula	Equation	Used in
Depletion width (10)	Charge balance	$N_a x_p = N_d x_n$	20-21 2(c)
	Depletion width	$W = x_n + x_p = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$	20-21 2(c)
	Each side	$x_n = W \frac{N_a}{N_a + N_d}$, $x_p = W \frac{N_d}{N_a + N_d}$	20-21 2(c)
	Peak field	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$	20-21 6(d)
Band diagram (5)	**Fermi potentials**	$\phi_{Fp} = V_t \ln \frac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t \ln \frac{N_d}{n_i}$	20-21 3(b)
	Built-in potential	$V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t \ln \frac{N_a N_d}{n_i^2}$	20-21 3(b), 20-21 6(d)
Biased junction (10)	**Law of the junction**	$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$	20-21 4(d)
	Majority carriers at the edge	$n_n = N_d + \delta p_n$	20-21 4(d)

Symbols:

x_n, x_p = depletion widths on the n and p sides ↩

Questions:

No.	Paper	Marks	Type	Question
Depletion width				
1	20-21 2(c)	0	derivation	Using the expression for V_{bi}, find the width of the depletion region.
2	20-21 6(d)	10	numerical	A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ *[sic: should be* $\epsilon_r = 11.7$*]*)
Band diagram				
3	20-21 3(b)	5	theory	Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}.
Biased junction				
4	20-21 4(d)	10	numerical	Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} *[sic: should be* 1×10^{16} *cm*$^{-3}$*]*. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side.

4.6. Tunnel diode:

5 marks, 1 question, in 1 of 1 papers. Book: Neamen Ch. 8, 15.

Most often theory (1 of 1). Subtopics: Operation.

Questions:

No.	Paper	Marks	Type	Question
Operation				
1	20-21 5(c)	5	theory	Explain the working principle of Tunnel diode. Draw its characteristics curve.

4.7. MOS structure:

15 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 10.

Most often theory (3 of 3). Subtopics: Band diagrams; Double gate; Inversion.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (5)	**Fermi potential, p substrate**	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	20-21 4(b)
Inversion (5)	**Threshold inversion**	$\phi_s = 2\phi_{fp}$	20-21 6(c)
	Maximum depletion width	$x_{dT} = \left[\frac{4\epsilon_s \phi_{fp}}{eN_a}\right]^{1/2}$	20-21 6(c)

Symbols:

ϕ_s = surface potential (band bending) ↩

Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 4(b)	5	theory	Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed.
Double gate				
2	20-21 6(a)	5	theory	Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS?
Inversion				
3	20-21 6(c)	5	theory	Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram.

4.8. Bipolar transistor:

10 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 12.

Most often theory (2 of 2). Subtopics: Operating modes; Minority distribution.

Formulas:

Subtopic	Formula	Equation	Used in
Operating modes (5)	**Equilibrium minority carriers**	$p_{E0} = \frac{n_i^2}{N_E}$, $n_{B0} = \frac{n_i^2}{N_B}$, $p_{C0} = \frac{n_i^2}{N_C}$	20-21 4(c)
	Four modes	active: $V_{BE} > 0$, $V_{CB} > 0$; saturation: $V_{BE} > 0$, $V_{CB} < 0$; cutoff: $V_{BE} < 0$, $V_{CB} > 0$; inverse: $V_{BE} < 0$, $V_{CB} < 0$	20-21 4(c)
Minority distribution (5)	**Base edges**	$n_B(0) = n_{B0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0} \exp\left(\frac{V_{BC}}{V_t}\right)$	20-21 5(b)
	Emitter and collector edges	$p_E(0) = p_{E0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $p_C(0) = p_{C0} \exp\left(\frac{V_{BC}}{V_t}\right)$	20-21 5(b)

Symbols:

N_E, N_B, N_C = emitter, base, collector doping ↩ x_B = neutral base width ↩

Questions:

No.	Paper	Marks	Type	Question
Operating modes				
1	20-21 4(c)	5	theory	Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities.
Minority distribution				
2	20-21 5(b)	5	theory	For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector?

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Semiconductor in equilibrium				
1	20-21 1(a)	5	Fermi level	$E_F - E_{Fi} = kT\ln(N_d/n_i)$ in n-type, $E_{Fi} - E_F = kT\ln(N_a/n_i)$ in p-type; on a log doping axis both are straight lines of 0.06 eV per decade (0.23, 0.35, 0.47, 0.53 eV at 10^{14} to 10^{19} cm^{-3}); E_F moves towards E_c with donors and E_v with acceptors, reaching the band edge at degeneracy.
2	20-21 2(a)	5	Compensation	Both donors and acceptors in one region; donor electrons fall into the acceptor states, so donors are ionized (+) and acceptors filled (−). For $N_d > N_a$: n-type, $n_0 \approx N_d - N_a$, E_F above E_{Fi}; for $N_a > N_d$: p-type; for $N_a = N_d$: behaves intrinsic.
Carrier transport				
3	20-21 3(c)	5	Mobility	Drift mobility $\mu = \lvert v_d\rvert/E = e\tau/m^*$. Scattering by lattice vibrations ($\mu_L \propto T^{-3/2}$) and ionized impurities ($\mu_I \propto T^{3/2}/N_I$), combined by $1/\mu = 1/\mu_L + 1/\mu_I$. The mean free time τ (about 10^{-13} s) sets the mobility and so the conductivity.
4	20-21 4(a)	5	Mobility	Lattice scattering grows with T, so $\mu_L \propto T^{-3/2}$ falls; impurity scattering weakens as carriers get faster, so $\mu_I \propto T^{3/2}/N_I$ rises; by Matthiessen's rule impurity scattering dominates at low T (mobility rises), lattice at high T (mobility falls); heavier doping lowers the curve.
Excess carriers				
5	20-21 1(b)	5	Minority carriers	Minority carrier: the less numerous carrier ($p_{n0} = n_i^2/N_d$). It diffuses because random thermal motion in a concentration gradient gives a net flow from high to low; injected minority carriers recombine as they spread, so their current is diffusion current. Poisson: $d^2\phi/dx^2 = -\rho/\epsilon_s$ links the space charge to the field and potential, giving E_{max}, V_{bi} and W of a junction.
6	20-21 5(a)	5	Carrier pulse	For n-type at low injection the ambipolar equation uses D_p, μ_p, τ_{p0}: $D_p\,\partial^2\delta p/\partial x^2 - \mu_p E_0\,\partial\delta p/\partial x - \delta p/\tau_{p0} = \partial\delta p/\partial t$. A pulse at $x = 0$ becomes a Gaussian that spreads, shrinks by recombination and drifts in the direction of E_0 at $\mu_p E_0$.
PN junction				
7	20-21 3(b)	5	Band diagram	E_F flat through the device; on the P side E_F is $e\phi_{Fp} = kT\ln(N_a/n_i)$ below E_{Fi}, on the N side $e\lvert\phi_{Fn}\rvert = kT\ln(N_d/n_i)$ above it; the bands are higher on the P side by eV_{bi} and bend only inside the space charge region; $V_{bi} = \lvert\phi_{Fp}\rvert + \lvert\phi_{Fn}\rvert$.
Tunnel diode				
8	20-21 5(c)	5	Operation	Both sides degenerately doped, so E_F lies inside the bands and the depletion layer is only 5 to 17 nm thick; electrons tunnel where filled states face empty ones; the current rises to a peak I_p, falls to a valley I_v (negative resistance), then rises as normal diode current; reverse bias tunnels freely.
MOS structure				
9	20-21 4(b)	5	Band diagrams	MOS on p-type Si. Accumulation ($V_G < 0$): bands bend up, holes gather at the surface. Depletion ($0 < V_G < V_T$): bands bend down, acceptor ions uncovered over x_d. Inversion ($V_G > V_T$): E_{Fi} at the surface falls below E_F, electrons form an n-type channel, $\phi_s = 2\phi_{fp}$ at threshold.
10	20-21 6(a)	5	Double gate	A thin (undoped) silicon body between two thin gate oxides with a gate on each side, n$^+$ source and drain (FinFET is its practical form). Benefits: better gate control of the channel, much smaller short-channel effects (DIBL, V_T roll-off), near-ideal subthreshold swing, low leakage, more drive current, no doping fluctuation.

No.	Paper	Marks	Subtopic	Points to cover
11	20-21 6(c)	5	Inversion	Positive gate voltage on a p-type body repels holes and bends the bands down by $e\phi_s$: depletion for $\phi_s < \phi_{fp}$, intrinsic surface at $\phi_s = \phi_{fp}$, n-type inversion layer once E_{Fi} falls below E_F; threshold at $\phi_s = 2\phi_{fp}$ ($n_s = N_a$), depletion width at its maximum x_{dT}.
Bipolar transistor				
12	20-21 4(c)	5	Operating modes	Forward active ($V_{BE} > 0$, $V_{CB} > 0$), saturation (both junctions forward), cutoff (both reverse), inverse active; a forward junction raises the minority edge value by e^{V/V_t}, a reverse one takes it to 0; draw the base, emitter and collector minority profiles for each mode with the edge values.
13	20-21 5(b)	5	Minority distribution	Forward active: base electrons fall almost linearly from $n_{B0}e^{V_{BE}/V_t}$ at the emitter edge to about 0 at the collector edge; emitter holes fall from $p_{E0}e^{V_{BE}/V_t}$ to p_{E0}; collector holes rise from about 0 to p_{C0}.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Semiconductor in equilibrium				
1	20-21 3(a)	5	Carrier concentrations	Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by n_0: quadratic; quadratic formula; keep the positive root; for p_0 put $n_0 = n_i^2/p_0$ instead.
2	20-21 6(b)	5	Ionization	Donor occupation with the factor $\frac{1}{2}$; non-degenerate: $n_d \approx 2N_d e^{-(E_d-E_F)/kT}$; $n_0 = N_c e^{-(E_c-E_F)/kT}$; the ratio cancels E_F; fraction $1/[1 + (N_c/2N_d)e^{-(E_c-E_d)/kT}]$; 0.4 % for P in Si at 10^{16} cm^{-3}, 300 K.
Carrier transport				
3	20-21 2(b)	5	Einstein relation	Graded n-type sample in equilibrium: $J_n = 0$; $n = N_c e^{-(E_c-E_F)/kT}$ with E_F flat; chain rule gives $dn/dx = -(n/kT)dE_c/dx$; $dE_c/dx = eE_x$; substitute: $D_n/\mu_n = kT/e$; repeat for holes.
PN junction				
4	20-21 2(c)	0	Depletion width	Charge density $-eN_a$, $+eN_d$; integrate Poisson once with $E = 0$ at both edges; continuity gives $N_a x_p = N_d x_n$; integrate again for the potential; $V_{bi} = \phi(x_n)$; substitute $x_p = N_d x_n/N_a$; solve for x_n, x_p; add for W.

7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Quantum basics					
1	20-21 1(c)	5	Density of states	Integrate the density of states $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$ from E_c to $E_c + kT$: $N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$.	$N = 2.13 \times 10^{19}$ cm^{-3}
2	20-21 2(d)	10	Fermi probability	(i) Set $f_F = 1/[1 + e^{(E-E_F)/kT}] = 0.1$ and solve: $E - E_F = kT \ln 9$. (ii) $n_0 = N_c \exp[-(E_c - E_F)/kT]$.	(i) $2.2\,kT$ (0.0569 eV); (ii) 1.16×10^{11} cm^{-3}

No.	Paper	Marks	Subtopic	Method	Answer
Semiconductor in equilibrium					
3	20-21 1(d)	10	Ionization	$n_0 = N_d$, $p_0 = n_i^2/n_0$; $E_c - E_F = kT\ln(N_c/n_0)$, $E_F - E_v = E_g - (E_c - E_F)$; fraction on donors $1/[1 + (N_c/2N_d)e^{-(E_c-E_d)/kT}]$.	$n_0 = 10^{15}$, $p_0 = 2.25 \times 10^5$ cm^{-3}; $E_c - E_F = 0.2652$ eV, $E_F - E_v = 0.8548$ eV; 99.95 % ionized
Carrier transport					
4	20-21 5(d)	10	Hall effect	$V_H = V_y$ from the figure; its sign gives the carrier type; $n = -I_x B_z/(edV_H)$ with d the thickness along z; $\mu_n = I_x L/(enV_x W d)$.	$V_H = -6.25$ mV; n-type (electrons); $n = 1.0 \times 10^{16}$ cm^{-3}; $\mu_n = 833$ cm^2/V s
Excess carriers					
5	20-21 3(d)	10	Steady state	Low injection in p-type: ambipolar equation becomes $D_n\,\delta n'' - \delta n/\tau_{n0} = 0$; solution decaying for $x > 0$: $\delta n = \delta n_0 e^{-x/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$; solve $\delta n(x) = 10^{14}$ for x.	$\delta n = 5 \times 10^{15} e^{-x/L_n}$, $L_n = 45.8$ μm; 10^{14} cm^{-3} at 179.3 μm
PN junction					
6	20-21 4(d)	10	Biased junction	$p_{n0} = n_i^2/N_d$; law of the junction $p_n(x_n) = p_{n0}e^{eV_a/kT}$ with $V_a = +0.7$ and -0.7 V; $n_n = N_d + \delta p$ (neutrality); check low injection.	$p_{n0} = 2.25 \times 10^4$; forward: $p_n = 1.23 \times 10^{16}$, $n_n = 2.23 \times 10^{16}$ cm^{-3} (low injection violated); reverse: $p_n \approx 0$, $n_n \approx 10^{16}$ cm^{-3}
7	20-21 6(d)	10	Depletion width	$V_{bi} = V_t \ln\!\left(N_a N_d/n_i^2\right)$; $W = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $x_n = WN_a/(N_a + N_d)$; $E_{max} = -eN_d x_n/\epsilon_s$.	$V_{bi} = 0.718$ V; $W = 0.452$ μm; $E_{max} = -3.18 \times 10^4$ V/cm

8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Quantum basics			
1	20-21 1(c)	Graph	density of states with the shaded strip
2	20-21 2(d)	Graph	Fermi-Dirac function with the 10 % point
Semiconductor in equilibrium			
3	20-21 1(a)	Graph	Fermi-Dirac function at three temperatures
4	20-21 1(a)	Diagram	band diagrams of intrinsic, n-type and p-type Si
5	20-21 1(a)	Graph	Fermi level against doping at 300 K
6	20-21 1(d)	Diagram	band diagram of the As-doped sample
7	20-21 1(d)	Graph	ionized fraction and Fermi level against temperature
8	20-21 2(a)	Diagram	band diagrams of a compensated semiconductor for three cases
9	20-21 2(a)	Graph	Fermi level against acceptor concentration
10	20-21 3(a)	Diagram	charge balance in a compensated semiconductor
11	20-21 6(b)	Diagram	donor states at 0 K, low temperature and 300 K
12	20-21 6(b)	Graph	n_0/N_d against temperature
Carrier transport			
13	20-21 2(b)	Diagram	graded doping profile and its equilibrium band diagram with the induced field

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No.	Paper	Kind	Diagram
14	20-21 3(c)	Diagram	electron path with and without a field
15	20-21 3(c)	Diagram	lattice and ionized impurity scattering
16	20-21 4(a)	Graph	mobility against temperature, log-log
17	20-21 5(d)	Diagram	Hall bar with the force balance
18	20-21 5(d)	Diagram	carriers piling up in p-type and n-type bars
Excess carriers			
19	20-21 1(b)	Diagram	diffusion of carriers down a concentration gradient
20	20-21 1(b)	Graph	charge density, field and potential of a PN junction
21	20-21 3(d)	Graph	excess electrons decaying into the bar
22	20-21 3(d)	Graph	total electron and hole concentrations on a log scale
23	20-21 5(a)	Diagram	excess electrons and holes held together by their field
24	20-21 5(a)	Graph	excess carrier pulse at three times, with and without a field
PN junction			
25	20-21 2(c)	Diagram	PN junction with its space charge region
26	20-21 2(c)	Graph	charge density, field and potential across an abrupt junction
27	20-21 3(b)	Diagram	p and n band diagrams before contact
28	20-21 3(b)	Diagram	equilibrium band diagram of a PN junction with all labels
29	20-21 4(d)	Diagram	band diagrams under forward and reverse bias
30	20-21 4(d)	Graph	hole and electron concentrations on the n side
31	20-21 6(d)	Graph	charge density, field and potential across the junction
Tunnel diode			
32	20-21 5(c)	Diagram	electron wave at a thick and a thin barrier
33	20-21 5(c)	Diagram	tunnel diode band diagrams at several biases
34	20-21 5(c)	Graph	tunnel diode I-V curve
MOS structure			
35	20-21 4(b)	Diagram	MOS capacitor and n-channel MOSFET cross sections
36	20-21 4(b)	Diagram	charges in accumulation, depletion and inversion
37	20-21 4(b)	Diagram	band diagrams in the three modes
38	20-21 6(a)	Diagram	single-gate, planar double-gate and FinFET schematics
39	20-21 6(a)	Graph	subthreshold current of single and double gate
40	20-21 6(c)	Diagram	band diagrams of a p-type MOS structure from depletion to inversion
41	20-21 6(c)	Diagram	charge distribution at threshold inversion
Bipolar transistor			
42	20-21 4(c)	Diagram	NPN structure and the four modes on the V_{BE}–V_{CB} plane
43	20-21 4(c)	Graph	minority carrier distributions in the four modes
44	20-21 5(b)	Diagram	NPN in forward-active mode with depletion layers
45	20-21 5(b)	Graph	minority carrier concentrations in emitter, base and collector

9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Semiconductor in equilibrium		
Complete ionization vs freeze-out	every donor gives its electron, $n_0 = N_d$ vs electrons fall back to donors at low T, $n_0 < N_d$	20-21 6(b)
Carrier transport		
Drift vs diffusion current	driven by a field, $en\mu E$ vs driven by a gradient, $eD\,dn/dx$; linked by $D/\mu = kT/e$	20-21 2(b)

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Pair	Key difference	Asked in
Lattice vs ionized impurity scattering	grows with temperature, $\mu_L \propto T^{-3/2}$ vs weakens with temperature, $\mu_I \propto T^{3/2}/N_I$	20-21 3(c)
Excess carriers		
Minority vs majority carriers	fewer, set by n_i^2/N, move by diffusion vs more, set by the doping, carry drift current	20-21 1(b)
MOS structure		
Accumulation vs depletion vs inversion	majority carriers gather vs carriers pushed away, ions uncovered vs minority carriers form a channel	20-21 4(b)
Single-gate vs double-gate MOSFET	gate controls one surface, short-channel effects vs gate on both sides, better control, near-ideal swing	20-21 6(a)

10. Repeat tracker:

No repeats among these papers.

11. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Quantum basics	·	·	2
Semiconductor in equilibrium	2	2	1
Carrier transport	2	1	1
Excess carriers	2	·	1
PN junction	1	1	2
Tunnel diode	1	·	·
MOS structure	3	·	·
Bipolar transistor	2	·	·

12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Semiconductor in equilibrium	30	21 %
2	Carrier transport	25	38 %
3	PN junction	25	55 %
4	Excess carriers	20	69 %
5	Quantum basics	15	79 %
6	MOS structure	15	90 %
7	Bipolar transistor	10	97 %
8	Tunnel diode	5	100 %

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